7.5 Enthalpy (H)

The enthalpy (H), is a thermodynamic property of a system, is defined as the sum of the internal energy (*U*) of a system and the product of pressure and volume of the system. That is,

$H = U + PV$ ----------(7.8)

It reflects the capacity to do mechanical work and the capacity to release heat by the system. When a process occurs at constant pressure, the heat involved (either released or absorbed) is equal to the change in enthalpy.

Enthalpy is a state function which depends entirely on the state functions T, P and U. Enthalpy is usually expressed as the change in enthalpy (ΔH) for a process between initial and final states at constant pressure.

$\Delta H = \Delta U + P\Delta V$ ----------(7.9)

The change in enthalpy (*ΔH)* is equal to the heat supplied at the constant pressure to a system (as long as the system does no additional work).

$\Delta H = q_p$

In an endothermic reaction heat is absorbed by the system from the surroundings that is $q>0$ (positive). Therefore, ΔH is also positive. In an exothermic reaction heat is evolved by the system to the surroundings that is, $q<0$ (negative). If q is negative, then ΔH will also be negative.

7.5.1 Relation between enthalpy `H' and internal energy `U'

When the system at constant pressure undergoes changes from an initial state with H_1, U_1 and V_1 to a final state with H_2, U_2 and V_2 the change in enthalpy ΔH, can be calculated as follows:

$H=U + PV$

In the initial state

$$H_1=U_1 + PV_1 \text{-----(7.10)}$$

In the final state

$$H_2 = U_2 + PV_2 \text{----(7.11)}$$

change in enthalpy is (7.11) - (7.10)

$$(H_2-H_1) = (U_2-U_1) + P(V_2-V_1)$$

$$\Delta H=\Delta U + P\Delta V \text{ --------- (7.12)}$$

As per first law of thermodynamics,

$\Delta U = q+w$

Equation 7.12 becomes

$\Delta H = q + w + P\Delta V$

$w= -P\Delta V$

$\Delta H = q_p - P\Delta V + P\Delta V$

$\Delta H = q_p$ ---------- (7.13)

q_p – is the heat absorbed at constant pressure and is considered as heat content.

Consider a closed system of gases which are chemically reacting to form gaseous products at constant temperature

11th Std Chemistry 186-228.indd 197 23-12-2021 15:33:47